Chapter 6 Part I – Cost Behavior

Shift in Focus

Away from GAAP	Product vs. Period Cost
	Manufacturing vs. Non-manufacturing Cost
	Allocation of indirect costs
Toward Internal Decision Making	Variable versus Fixed Costs (Cost Behavior)
	Cost-Volume-Profit
	Incremental Analysis

Learning Objectives

Describe key characteristics and graphs of various cost behaviors

Use cost equations to express and predict costs

Understanding cost behavior

Understanding cost behavior is the key to many managerial decisions.

How does total cost change when something else changes, such as:

- Producing more units
- Serving more customers
- Dropping a product line
- Outsourcing part of a process

Cost Behavior

Most common cost behaviors:

- Variable costs
- Fixed costs
- Mixed costs

Variable Costs

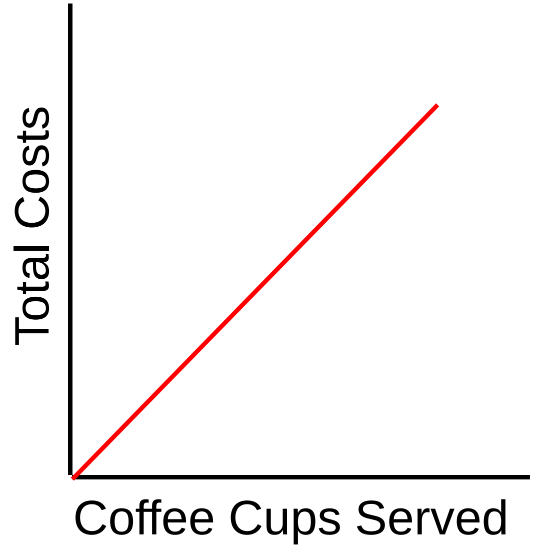
Costs that are incurred for every unit of volume.

Total variable costs change in direct proportion to changes in volume.

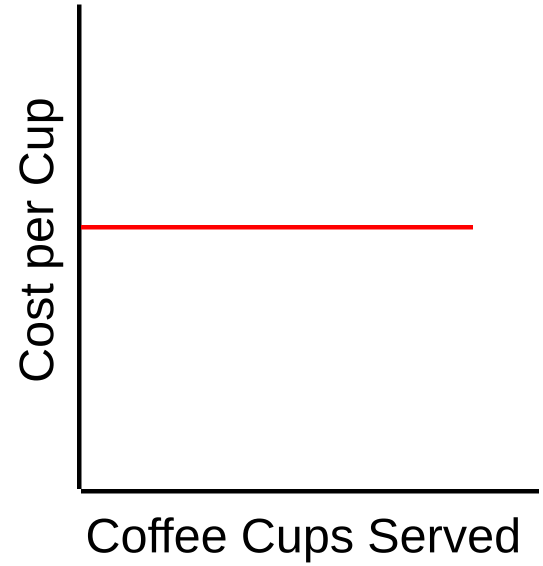
Graphs always begin at the origin.

Slope represents the variable cost per unit of activity.

Variable Costs Graph



Total variable costs increase as activity increases.



Cost per cup is constant as activity increases.

Cost Equation—Variable Costs

Mathematical equation for a straight line, to express how a cost behaves

Variable cost line: **y = vx**, where

- y = total variable cost
- v = variable cost per unit of activity
- x = volume of activity

Variable cost *per unit* remains constant

Summarize : Key Characteristics of Variable Costs

- *Total* variable costs change in *direct proportion* to changes in volume
- The *variable cost per unit of activity* (v) remains constant and is the slope of the variable cost line
- Total variable cost graphs always begin at the origin (if volume is zero, total variable costs are zero)
- Total variable costs can be expressed as follows:

  $$y = vx$$

  where,

  y = total variable cost

  v = variable cost per unit of activity

  x = volume of activity

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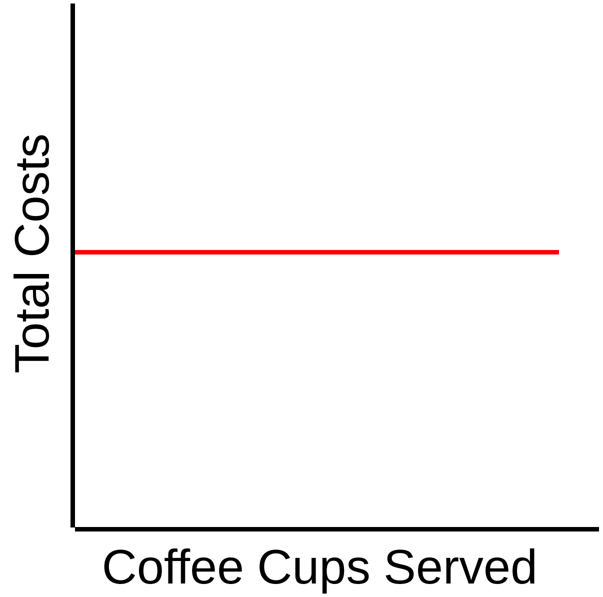
Fixed Costs

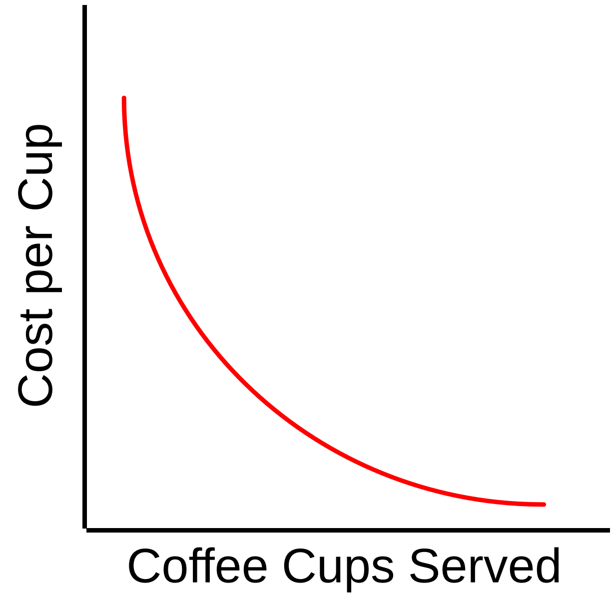
Costs that do ***not*** change in total despite changes in volume

Committed fixed costs	No control over these Property taxes, insurance, depreciation, lease payments, etc
Discretionary fixed costs	Relatively more control Manager salaries, advertising expense, etc.
Graph—flat line that intersects y-axis	

Fixed costs



Total fixed costs remain constant as activity increases.



Cost per cup declines as activity increases.

Cost Equation—Fixed Costs

Fixed cost *per unit* is inversely proportionate to volume.

y = f, where

- y = total fixed cost
- f = fixed amount over a period of time

Summarize - Key Characteristics of Fixed Costs

- *Total* fixed costs stay *constant* over a wide range of volume
- Fixed costs *per unit of activity* vary *inversely* in proportion to changes in volume:
  - Fixed cost per unit of activity *increases* when volume *decreases* *(If volume is cut in half, the fixed cost per unit will double)*
  - Fixed cost per unit of activity *decreases* when volume *increases* *(If volume doubles, the fixed cost per unit will be cut in half)*
- Total fixed cost graphs are always flat lines with no slope that intersect the y-axis at a level equal to total fixed costs
- Total fixed costs can be expressed as $y = f$

  where,

  y = total fixed cost

  f = fixed cost over a given period of time

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Mixed Costs

Contain both variable and fixed cost components

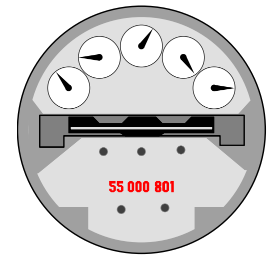
Electric Utility bill – cost of wires for transmission → fixed

Electric Utility bill – consumption of electricity → variable

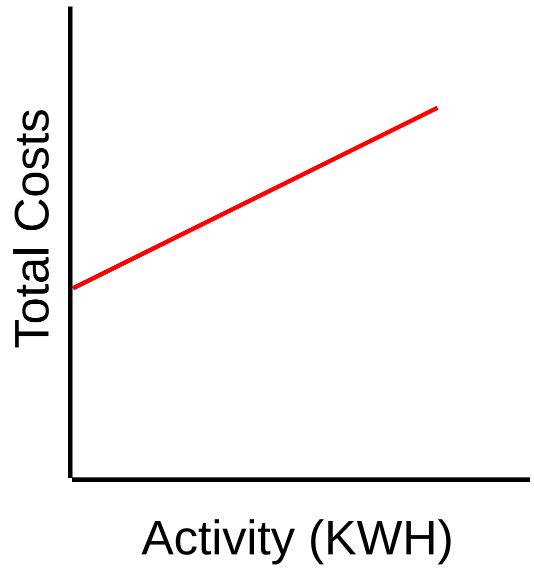
Increases with volume

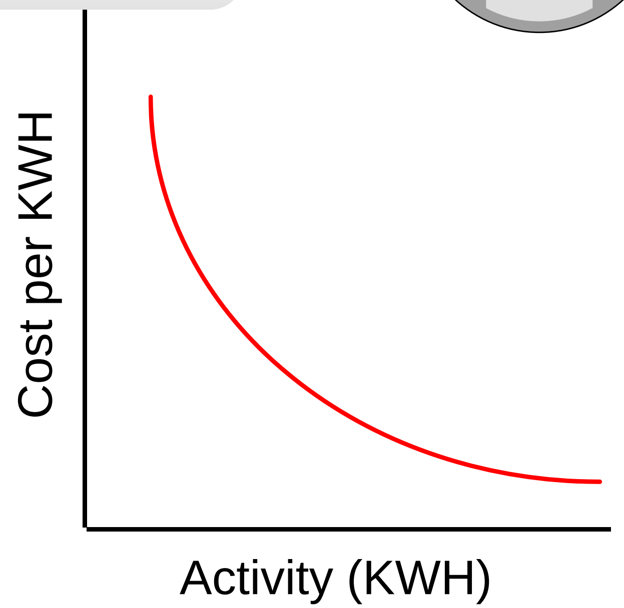
Does *not* start at the origin

Mixed costs



Example:
Electric
Utility
Bill



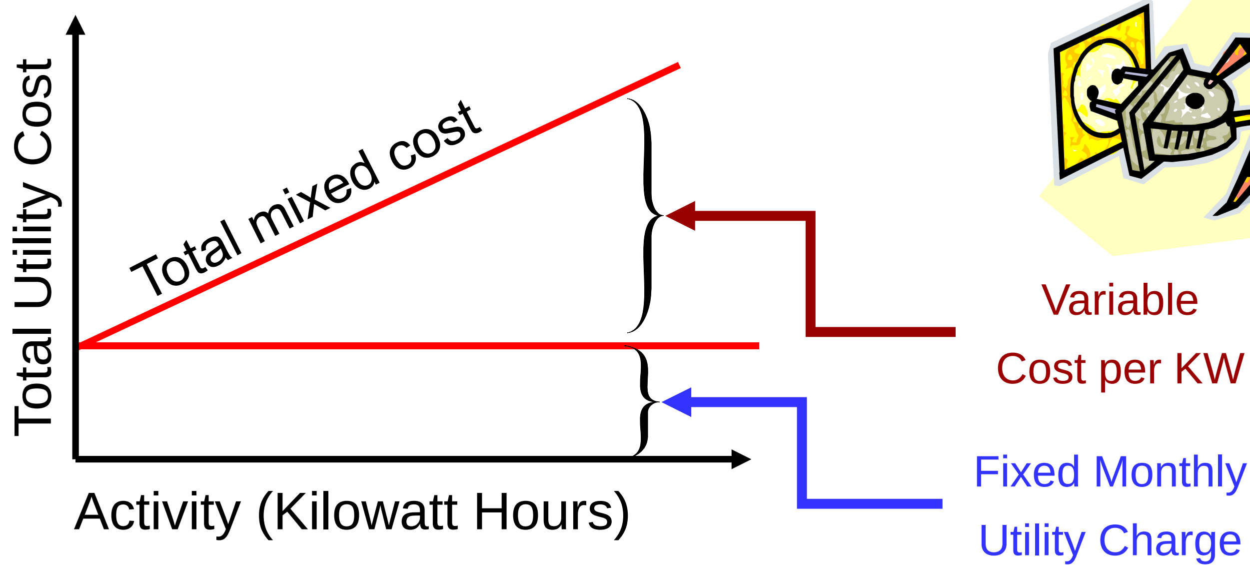


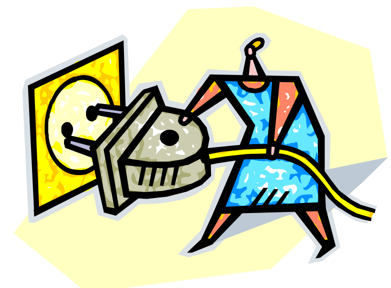
Total mixed costs increase as activity increases.

Per unit mixed costs decrease as activity increases.

Mixed costs

Mixed costs contain a fixed portion that is incurred even when facility is unused, and a variable portion that increases with usage. Utilities typically behave in this manner.





Cost Equation—Mixed Costs

Total mixed costs increase when volume increases, but ***not*** in direct proportion.

y = vx + f,
where

- y = total mixed costs
- v = variable cost per unit
- x = number of variable units
- f = fixed cost component

Key Characteristics of Mixed Costs

- *Total* mixed costs increase as volume increases because of the variable cost component
- Mixed costs *per unit* decrease as volume increases because of the fixed cost component
- Total mixed costs graphs slope upward but do *not* begin at the origin—they intersect the y-axis at the level of fixed costs
- Total mixed costs can be expressed as a *combination* of the variable and fixed cost equations:

Total mixed costs = variable cost component + fixed cost component

$$y = vx + f$$

where,

y = total mixed costs

v = variable cost per unit of activity (slope)

x = volume of activity

f = fixed cost over a given period of time (vertical intercept)

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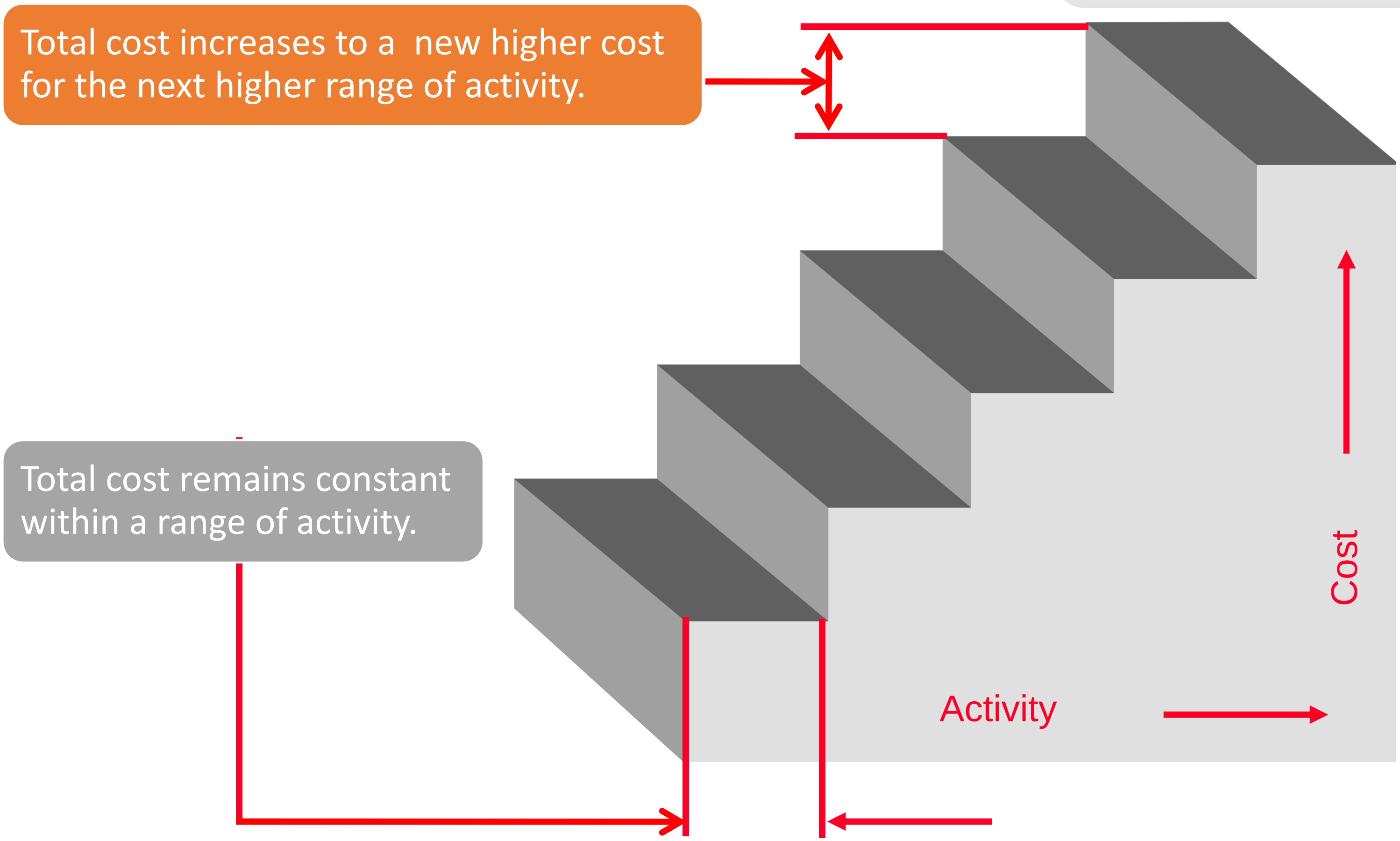
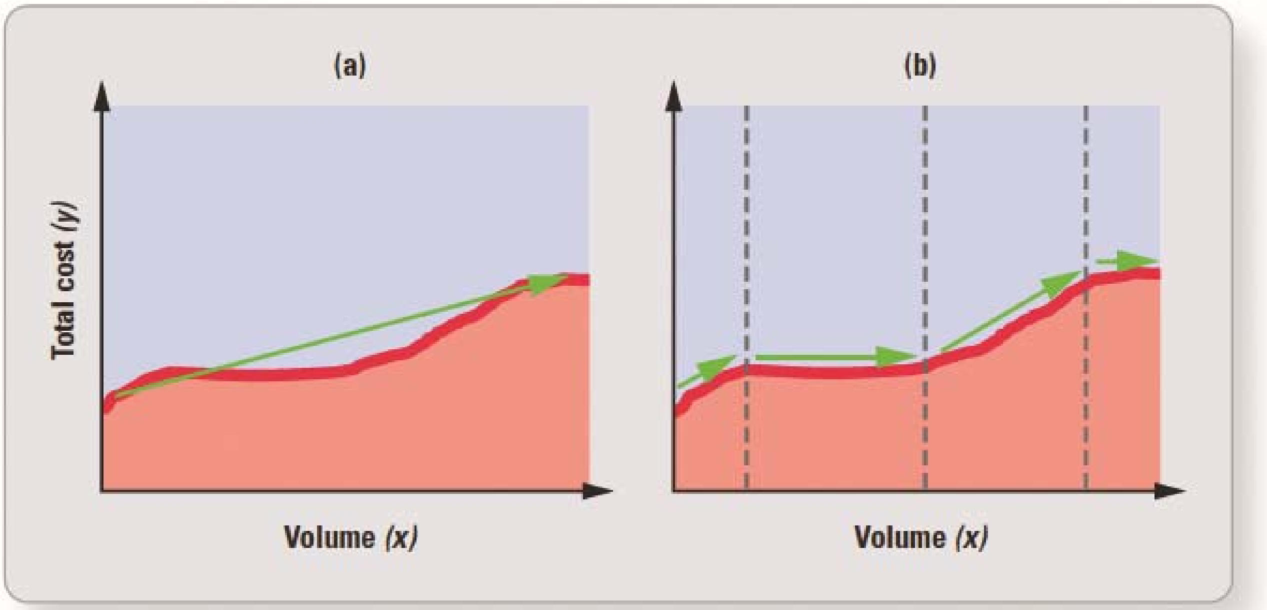
Step costs

Example: Child care centers where ratio of teachers to students is regulated.



Other Cost Behaviors—Curvilinear Costs

- **Curvilinear costs:** not linear, do not fit into any neat pattern



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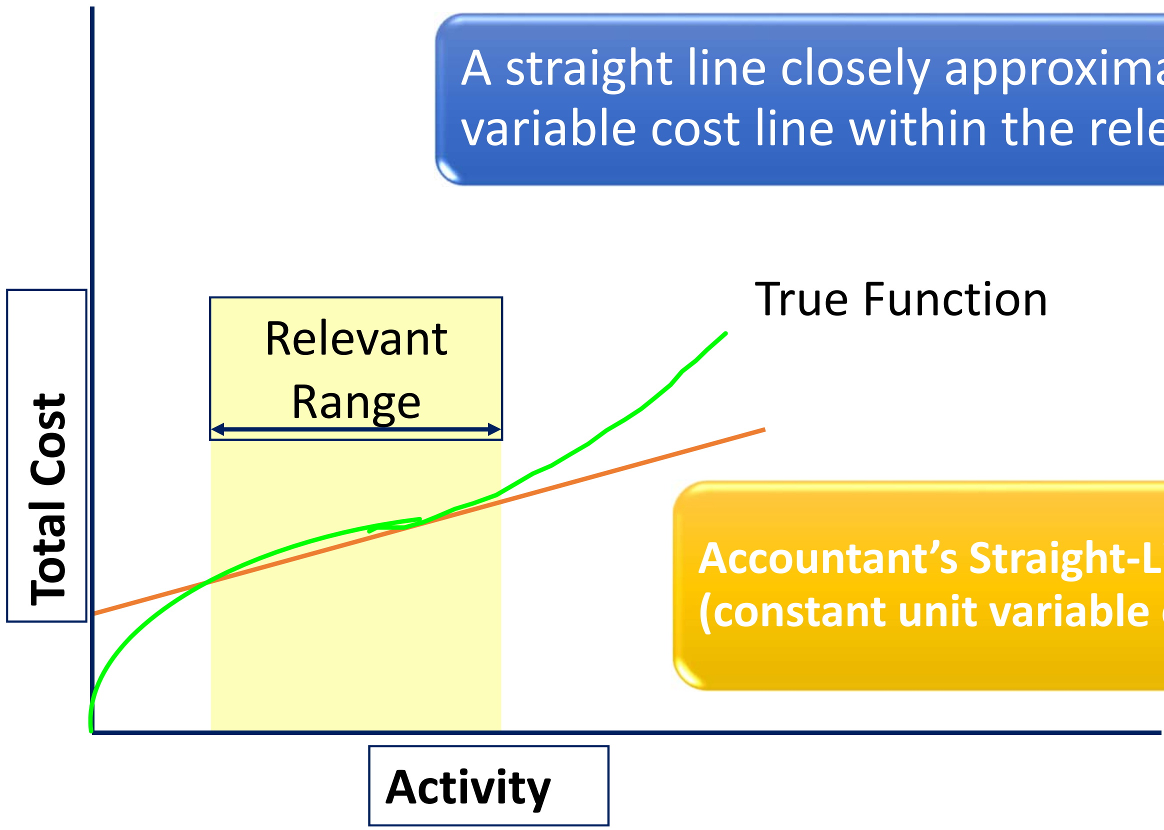
Constraints

Relatively short time horizon	Generally less than 1 year.
Relevant range	A limited range of activity over which we expect our assumptions about cost behavior to hold true.
	E.g. We typically assume that the relationship between total cost and activity is LINEAR!

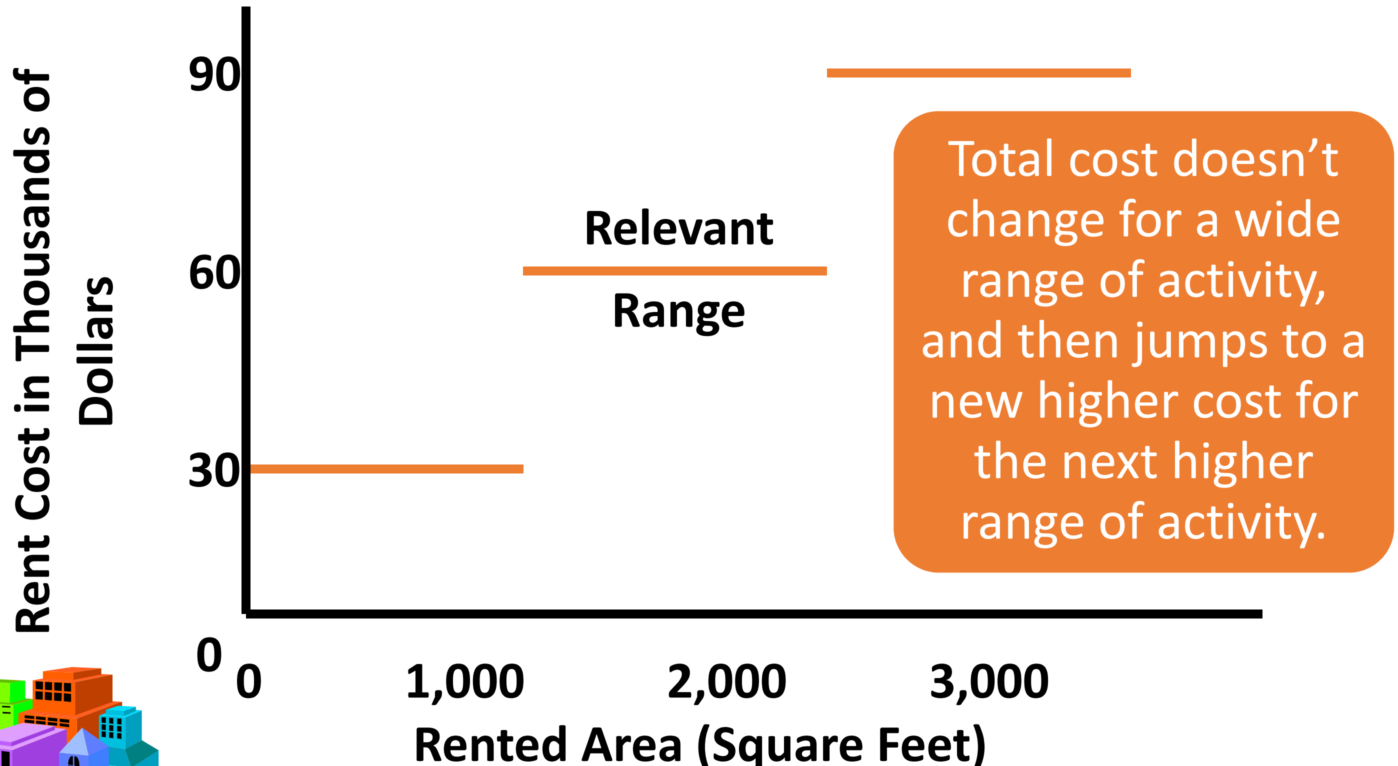
The Linearity Assumption and the Relevant Range

A straight line closely approximates a curvilinear variable cost line within the relevant range.



Accountant's Straight-Line Approximation (constant unit variable cost)

Fixed Costs and Relevant Range



Relevant Range

The band of volume where the following remain constant:

- Total fixed costs
- Variable cost per unit